



Valuing avoided morbidity using meta-regression analysis: what can health status measures and QALYs tell us about WTP?

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Summary

Many economists argue that willingness-to-pay (WTP) measures are most appropriate for assessing the welfare effects of health changes. Nevertheless, the health evaluation literature is still dominated by studies estimating nonmonetary health status measures (HSMs), which are often used to assess changes in quality-adjusted life years (QALYs). Using meta-regression analysis, this paper combines results from both WTP and HSM studies applied to acute morbidity, and it tests whether a systematic relationship exists between HSM and WTP estimates. We analyze over 230 WTP estimates from 17 different studies and find evidence that QALY-based estimates of illness severity – as measured by the Quality of Well-Being (QWB) Scale – are significant factors in explaining variation in WTP, as are changes in the duration of illness and the average income and age of the study populations. In addition, we test and reject the assumption of a constant WTP per QALY gain. We also demonstrate how the estimated meta-regression equations can serve as benefit transfer functions for policy analysis. By specifying the change in duration and severity of the acute illness and the characteristics of the affected population, we apply the regression functions to predict average WTP per case avoided. Copyright © 2006 John Wiley & Sons, Ltd.

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Introduction

Economic evaluation methods – in particular cost–benefit analysis (CBA) and cost-effectiveness analysis (CEA) – are now widely used to analyze and set priorities for public health and safety programs. Whereas CEA continues to be the method of choice for evaluating health care programs [1], CBA is often required by governments for evaluating environmental, food safety, and other regulatory programs.

A primary difference between these two methods is the metric they use to evaluate changes in

health risks or outcomes. CBA requires that health changes be expressed in monetary terms. Although placing a dollar value on human health is often controversial, welfare economists generally agree that willingness-to-pay (WTP) provides a conceptually appropriate monetary measure of health benefits. In contrast, CEA stops short of monetizing health impacts and instead relies a wide variety of health-based effectiveness measures. Quality-adjusted life years (QALYs) in particular are now commonly used as a nonmonetary summary measure of effectiveness [2].

Because of CBAs prominent role in evaluating public health and safety policies, it is essential for

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policy analysts in these areas to be able to consistently estimate WTP for avoided illness. However, faced with inevitable time and resource constraints, analysts are typically limited in their ability to conduct primary health valuation research for specific policy initiatives. Instead, they rely on methods for transferring value information from existing studies to the policy setting being evaluated. In other words, they develop and apply 'benefit transfer' methods that make best use of the available empirical literature on health preferences. This literature can include QALY-based measures that, like WTP measures, are designed to represent individual preferences.

A critical issue for CBA in public health policy is therefore whether information from QALY studies can be systematically used and combined with WTP studies to develop effective benefit transfer methods. Although primarily designed for CEA, can QALY-based estimates be used to support CBA? This paper uses a specific type of meta-analysis – meta-regression analysis – to address this issue, as it applies to the valuation of avoided acute morbidity.

In several important respects, this study builds on a previous meta-regression analysis conducted by Johnson *et al.* [3]. In particular, we supplement the Johnson *et al.* data with estimates from 12 more recent studies, and we include and control for a number of additional study characteristics through meta-regression. The results of our analysis confirm the findings of Johnson *et al.* with a high level of statistical significance, and they also extend beyond Johnson *et al.* in several ways. In particular, we explicitly test the assumption of a constant WTP per QALY gain. This practice of placing a dollar value on a QALY has been used in several CBA applications [4,5]; however, our results contradict the assumptions underlying this approach.

We begin the next section with a description of meta-regression analysis, showing how it can serve as a tool to synthesize research findings, test hypotheses, and apply for policy analysis. In the following section, we describe a simple conceptual model of health valuation, and in the next section, we define a number of testable hypotheses based on this framework. Then the forthcoming section describes the data used in the analysis, and in the following section, we describe the results of these tests based on meta-regression analyses. Depending on the statistical significance and explanatory power of the meta-regressions, meta-analysis

results can also be used for prediction. The ability to reliably predict WTP is an essential criterion for conducting benefit transfer. Therefore, in this context, meta-regression equations can serve as benefit transfer functions. In the penultimate section, we demonstrate and evaluate this application of meta-analysis results. The last section concludes.

Analytical method: meta-regression analysis

Broadly speaking, meta-analysis refers to the practice of using statistical methods to synthesize the results from a well-defined class of empirical studies. Although it has primarily evolved and been applied in the area of health sciences, it is increasingly being used in social science research. Meta-regression analysis, which is a specific kind of meta-analysis, has particularly evolved in economics as a method for summarizing and analyzing empirical results [6].

Smith and Pattanayak [7] provide a recent summary and evaluation of how meta-regression analysis has specifically been used in nonmarket (including health) valuation. They argue that meta-analyses in this field have served three main purposes:

- research synthesis,
- hypothesis testing, and
- prediction.

In this paper, we use meta-regression analysis to address all three of these objectives.

As a method for synthesizing research findings, meta-analysis goes beyond most literature reviews by identifying quantitative measures that can be defined consistently across studies and providing statistical summaries of these measures. In this case, we gather and standardize information on individuals' WTP to avoid acute illness, and we summarize results across a range of studies. The selected studies and WTP estimates for this study are summarized in the 'Data' section and in Appendix A.

Meta-regression analysis expands the scope for hypothesis testing beyond traditional meta-analysis. It allows the analyst to control for and separately estimate the effects of different covariates on the measure of interest. In the context of nonmarket valuation, the core hypothesis of interest is

whether observed variation in WTP estimates is consistent with theory. For example, evidence that WTP varies significantly across studies with respect to average income and the size of the quality change [8] supports the validity and credibility of the research.

Conceptual framework for health valuation

To formalize the way in which individuals derive value from changes in health, one can begin with a simple conceptual framework that links an individual's private utility (U) with his or her health status (H). This simple single-period framework, which assumes that H is exogenously determined and that there is no uncertainty regarding H or other determinants of utility, is summarized in the following utility function:

$$U = U(H, X) \quad (1)$$

In this function, X represents a vector of other goods that contribute to utility. Individuals are assumed to maximize this utility function subject to the budget constraint $Y = PX$, where Y is exogenously determined income and P is a vector of prices corresponding to the X goods.

The indirect utility function associated with this maximization process can be written as follows:

$$V = V(H, Y, P) \quad (2)$$

The monetary value, or WTP, associated with an improvement in health from initial health, H^0 , to new health, H^1 , can be expressed by the compensating variation term (CV) in the following equation:

$$\begin{aligned} V(H^0, Y, P) &= V(H + \Delta H, Y - CV, P) \\ &= V(H^1, Y - CV, P) \end{aligned} \quad (3)$$

CV represents the reduction in income that would exactly offset the increase in utility resulting from health improvement, such that there would be no *net* gain in utility. Similarly, the WTP associated with *avoiding* a decline in health from H^0 to H^2 can be expressed by the equivalent variation term (EV) in the following equation:

$$\begin{aligned} V(H^0 - \Delta H, Y, P) &= V(H^2, Y, P) \\ &= V(H^0, Y - EV, P) \end{aligned} \quad (4)$$

EV represents the reduction in income that would reduce utility by exactly the same amount as the health decline.

Both Equations (3) and (4) can be rearranged to derive the following corresponding value functions:

$$CV = CV(H^0, H^1, P, Y) \quad (3')$$

$$EV = EV(H^0, H^2, P, Y) \quad (4')$$

These equations represent 'variation' or WTP functions with respect to a change in health status. They provide a conceptual starting point for constructing and estimating meta-analytic functions of WTP for avoided illness. Both CV and EV are considered to be conceptually correct private welfare measures for changes in health; however, even for equivalent gains or losses in health (i.e. holding ΔH constant), they will not necessarily have the same value. The potential differences between these two measures are described in more detail below. CV can also be expressed as the *increase* in income that would exactly offset the utility loss associated with a *decline* in health (i.e. an individual's minimum willingness to accept (WTA) compensation for a decline in health). Although much attention has been devoted in the literature to discrepancies between WTP and WTA measures [9,10], in practice relatively few studies have estimated WTA values for health changes.

Although the CV and EV measures in Equations (3) and (4) are appropriate private welfare measures for health improvements, it is important to emphasize that they will underestimate societal welfare if some costs of illness are externalized to others. As shown by, for example, Harrington and Portney [11], private WTP represented in Equations (3) and (4) should reflect and include individuals' own avoided (1) direct costs of illness, such as medical expenses, (2) indirect costs such as lost income or household production, and (3) pain and suffering. To the extent that portions of these costs are externalized to others, for example through public or private insurance or sick leave policies, they will not be reflected in individuals' private WTP to avoid illness or improve health. These costs, including pain and suffering, may also be externalized through *altruistic* values. That is, certain individuals may lose utility when others experience illness, and they may be willing to pay for the changes in others' health. In particular, parents are often willing to pay for improvements in their children's health. Although it is possible to

develop CV and EV measures that incorporate altruistic values for others' health [12,13], their inclusion in CBA is often problematic [14].

Hypotheses regarding determinants of WTP

Based on the logic represented in Equations (3') and (4'), we expect WTP estimates for health improvements to be primarily influenced by the magnitude of changes in health outcomes and the characteristics of the study population. From an empirical standpoint, we also expect value estimates to be influenced by the valuation method used. Below, we discuss how these determinants of health values can be measured and how they are expected to affect the magnitude of value estimates.

Duration and severity of health effects

Although it is well understood that health outcomes are inherently complex and multidimensional, for the purposes of economic evaluation it is generally necessary to develop simplified summary measures of health. Therefore, we characterize acute conditions in two primary dimensions: duration and severity.

The primary expectation is that larger reductions in the duration and/or severity of illness will generate larger value estimates (positive first derivatives). The test of this type of relationship (i.e. between WTP and the size of the 'commodity' being valued) is commonly referred to as a 'scope test' [8]. The primary challenge in testing for scope effects is therefore to define measures of duration and severity that can be applied consistently across the health effects being valued in the WTP studies.

Specifying a continuous measure of duration for acute conditions is relatively straightforward. For example, it can be expressed as the number of days during which one experiences the acute condition. However, it should be noted that in some instances the acute condition may be experienced as a single event lasting multiple days (e.g. an acute respiratory infection) and in other instances the acute condition may be experienced as multiple discrete events on separate days (e.g. angina or asthma attacks). In both cases, in this paper we use the

term duration to refer to the number of days with the acute condition.

Measures of severity, by contrast, are inherently more complex. Fortunately, economists, psychometricians, and other health experts have developed a number of 'generic' health status measures (HSMs) that can be used to characterize severity for a wide range of health outcomes. These measures typically define a set of possible discrete health states, by defining specific health dimensions (e.g. physical, mental, and emotional domains) and discrete levels of health for each dimension. In addition to these methods for categorizing health states, a number of survey-based scoring techniques have been developed for measuring and comparing the severity of health states. These techniques – in particular standard gamble (SG), time tradeoff (TTO), and visual analog scale (VAS) approaches – use preference elicitation methods to produce 'utility weights' for each possible health state (see, for example, [15] for a summary and evaluation of these scoring techniques). Typically these weights are constructed to vary between two extremes – 0 (immediate death) and 1 (perfect health).

Multiattribute utility scales (MAUSs) represent a specific type of HSM. They are of particular interest for quantifying severity of illness because they merge *categorization* of health states (HSM techniques) with *utility measurement* for health states. In other words, they use SG, TTO, or VAS approaches to derive preference-based utility weights/scores (usually on a 0–1 scale) for a wide range of well-defined health states. The three most widely used MAUSs are the Quality of Well-Being (QWB) Scale, the Health Utility Index (HUI), and the EuroQol. Table 1 compares some of the primary features of these three methods (see [16] for a more detailed summary of these methods). Each method has advantages; however, because of the way it specifies dimensions and levels of health, the QWB Scale can be most readily applied as a severity index for acute illness.

One of the distinct features of the QWB Scale is that it provides separate scores for each of the four health dimensions that it covers – symptoms, mobility, physical activity, and social activity. The 27 symptoms range in severity from eye irritation and shortness of breath to loss of consciousness and severe burns. The mobility, physical activity, and social activity scales each primarily include two levels of impairment, one involving modest restrictions and other involving severe restrictions

Table 1. Comparison of three commonly used multiattribute utility scales

MAUS	Health dimensions	Levels/ dimension	Total number of health states
QWB	3 functioning dimensions: mobility, physical activity, social activity + 27 symptoms/problems	3 2	1170
HUI-III	8 dimensions: vision, hearing, speech, ambulation, dexterity, emotion, cognition, pain/discomfort	5–6	972 000
EuroQol	5 dimensions: mobility, self-care, usual activity, pain/discomfort, anxiety/depression	3	243

Source: Adapted from [16].

(e.g. hospitalization and unable to perform normal activities). A crucial assumption in developing the composite (total) QWB score is that it is an additive combination of the four subcomponent scores. If each of these subcomponents contributes equally to utility, then, as measures of changes in severity, they should also have the same impact on WTP. Through a meta-analysis of WTP estimates, it is possible to test this restriction.

QALYs take health measurement one step further, by combining information on severity (typically from MAUS or related utility scores) and duration in a very specific way. According to the QALY framework, the health-related quality of life corresponding to any possible health state ($i = 1$ to N) can be represented by a single utility index or weight (q_i), generally varying between 0 (death) and 1 (perfect health), although in some instances weights less than 0 (worse than death) are also considered. These 'QALY weights' are typically derived from SG, TTO, or VAS scoring approaches or from MAUS methods that use these techniques. If T_i presents the number (or fraction) of life years spent in each health state, i , then the number of QALYs corresponding to a lifetime health profile can be expressed as

$$\text{QALY} = \sum_{i=1}^N q_i T_i \quad (5)$$

When QALYs are used as a representation of individual preferences, they imply that the utility (or disutility) associated with time spent in any health state is directly proportional to the duration of the health state. That QALYs are defined to be independent of income (and thus consumption) is in some instances viewed as an advantage because they treat health benefits equitably across any given population (i.e. without regard to individual

economic circumstances). However, independence from income also imposes strong restrictions for interpreting QALYs as a representation of preferences [17].

The assumptions underlying the QALY model also define hypotheses regarding how changes in severity and duration should affect values. Under the most restrictive set of assumptions where QALYs are interpreted as a utility function and consumption is constant over time [18], WTP should increase in direct proportion to the gain in QALYs. This implies the following expression:

$$\text{WTP} = \alpha(\Delta\text{QALY})^\beta \quad \text{where } \beta = 1 \quad (6)$$

In this expression, α can be interpreted as a constant value per QALY gain. In practice, a constant value per QALY or 'QALY valuation' approach has been frequently used (see, for example, [4,5]) for valuing national-level health changes. In most of these applications, the value per QALY is derived by annualizing empirical estimates of the value per statistical life (VSL) [18], resulting in values of roughly \$100 000 per QALY.

Maintaining the constant value per QALY assumption, if the QALY change is composed of a specific change in health-related quality of life (Δq) over a specific time period (Δt), then Equation (6) can be further decomposed into the following expression:

$$\text{WTP} = \alpha(\Delta q \Delta t)^\beta \quad \text{where } \beta = 1 \quad (7)$$

This formulation implies that the elasticities of WTP with respect to Δq and Δt are both equal to 1. Through a meta-analysis of WTP estimates, it is possible to test this hypothesis directly.

As shown by Hammitt [How much is a QALY worth? Utility functions for health and wealth. Unpublished manuscript, 2002.] and Klose [19], under somewhat less restrictive conditions, it is

possible to include QALYs (or QALY weights) in a utility theoretic framework without imposing assumptions that imply a constant WTP per QALY. In other words, QALY weights that are invariant to wealth and longevity can be derived from plausible preference structures that include both health and consumption. Under these conditions, WTP is systematically but not linearly related to QALY gains. In particular, under most plausible conditions, WTP per QALY gain is declining with respect to health status and the size of the QALY gain.

Study population characteristics

Values for health changes are also expected to depend importantly on the characteristics of the individual in question, in particular income, prices, age, and health status.

Income and wealth are primarily expected to have a positive effect on health values. First, individuals' WTP for health changes is constrained by their available budget; therefore, other things equal and as long as health is a normal good, it is expected that individuals with higher incomes will also have higher values for health changes. Second, to the extent that higher incomes imply higher opportunity costs of illness (e.g. through lost wages or household production), they create a price effect. A higher price of illness is expected to increase WTP for health. Higher prices of mitigating goods, such as medication and treatment, also affect the direct cost of illness and are expected to increase the WTP to avoid or reduce morbidity. It is important, however, to emphasize that these prices may be partially or totally covered by employers or health insurers, in which case the marginal effect of these prices on private WTP should be smaller.

An individual's age can also potentially affect his or her WTP for health improvements in a number of ways, but for avoided acute effects it is generally expected to have a positive effect. First, in general terms, age is negatively related to health status. To the extent that marginal utility for improved health status is declining ($\partial^2 V / \partial H^2 < 0$), and age serves as an indicator (inverse) of health status, a positive relationship between WTP and age is expected. Second, age may capture aspects of socio-economic status that are not captured by annual income measures. For example, elderly individuals may have higher accumulated wealth

relative to younger individuals with comparable income. If wealth measures are not adequately controlled for, then age may increase WTP for health improvements through a wealth effect.

Valuation method

The magnitude of value estimates for changes in health is also expected to depend on the way in which they are estimated. All the WTP estimates included in our analysis are based on stated preference (SP) methods (either the contingent valuation method (CVM) or conjoint analysis). Even so, there are a number of methodological differences that have the potential to influence WTP.

One potentially significant difference is the type of value elicitation format used – open-ended (OE) or dichotomous choice (DC). The relative advantages of these formats have been widely discussed and analyzed. Carson [20], for example, argues in favor of the DC format. He suggests that individuals will tend to understate their WTP with the OE format because respondents do not have the incentive and are not accustomed to conditions where they have to 'find' their maximum point. However, if respondents are answering questions strategically, OE responses may overstate or understate WTP [21]. Others have argued that the DC format is susceptible to 'yea-saying,' in which case respondents exhibit an inherent tendency to agree with questions [22–25]. Therefore, the expected direction of this effect is indeterminate, but meta-analysis offers an approach for testing whether systematic differences exist between DC- and OE-based estimates of health values.

Another potentially important difference has to do with how the health change (i.e. the 'commodity') is described. As previously discussed, the valuation context can be framed as either WTP to improve health (CV) or WTP to avoid a decline in health (EV). Even for equivalent sized gains or losses in health (i.e. holding ΔH constant), they will not necessarily have the same value. In particular, if the marginal utility of health decreases with H ($\partial^2 V / \partial H^2 < 0$) or if the marginal utility of income increases with H ($\partial^2 V / \partial Y \partial H > 0$), EV is generally expected to be greater than CV.

Nonetheless, for small changes in health, the two measures should be roughly equivalent. Large differences between CV and EV are possible, but they may also indicate deviations from the standard utility model, such as the loss aversion and

reference dependence models proposed by Tversky and Kahneman [26]. According to this alternative framework, individuals value changes with respect to a reference point (such as their status quo condition) and place a much higher (negative) weight on losses than they do on equivalent gains.

Data

As Smith and Pattanayak [7] note in their review of meta-analyses in nonmarket valuation, 'Synthesis requires the ability to define a common concept to be measured' (p. 274). The number of empirical health valuation studies has increased significantly over the last 10–15 years, but there is also significant heterogeneity in terms of the valuation approaches used and health effects evaluated across these studies. To define a meaningful common concept for meta-analysis, we used a similar approach to Johnson *et al.* [3].

In particular, we selected WTP values if

- they were estimated for well-defined acute health effects,
- they represented adults' private values for improving their own health,
- they were estimated using SP methods,
- the severity of the acute health effect could be expressed as and converted to a QWB score, and
- the change in duration of the acute effect could be quantified in terms of discrete days or episodes of illness.

We began by selecting the 53 observations included in the Johnson *et al.* study. These values were taken from five CV studies conducted in the United States in the late 1970s and 1980s. The studies were predominantly conducted for cardio-respiratory health effects associated with air pollution. We then conducted an extensive and detailed search and review of the health valuation literature to identify additional studies to include in the analysis. This search included other existing literature reviews [27–29] as well as several online databases (e.g. PubMed, ingenta, Econlit) using key words such as WTP, WTA, health value, health valuation, contingent valuation, and conjoint analysis. We also scanned the reference lists of these articles and consulted with other experts

in the field to identify additional candidate studies. Using these methods, we identified 183 additional value estimates (from 12 other studies), which expands the estimation sample by a factor of more than four. These additional studies were, for the most part, conducted after 1990, and they include research conducted both in the United States and in other countries. Appendix A provides more detailed descriptions of some of the key characteristics of these studies, including the valuation methods used, the health effects valued, the country studied, and the number of value estimates included in the meta-analysis.

As is commonly done in meta-regression analyses, we did not define overly restrictive quality criteria for including or excluding studies from the analysis. Rather, we followed the recommendation of Stanley [6] who writes 'when in doubt, it is best to err on the side of inclusion... Differences in quality, data, or methods do not provide a valid justification for omitting studies [from meta-analysis]' (p. 135). Excluding studies based on quality would require making judgments that might introduce more bias into the results than it would avoid. Moreover, to the extent possible, we have included explanatory variables in the analysis to control for methodological differences.

For the 236 selected values, Tables 2 and 3 describe and summarize the main variables that we used in the meta-analysis. WTPACUTE was the key variable of interest. It represents individuals' WTP to avoid or to reduce the number of days with a specific acute condition over the course of a year. All WTP estimates were converted to 2000 US dollars using the consumer price index (CPI), and, if they were originally measured in another currency, we first converted the estimates to dollars using the purchasing power parity (PPP) index. Most of the selected studies estimate and report average WTP values. If only median WTP values were reported, we included these estimates in WTPACUTE.

Following the approach used by Johnson *et al.*, we characterized the change in acute health outcomes associated with each WTP value in two main dimensions. First, we created the variable Δ DAYS to capture changes in the duration of the health effect. In most cases, this variable represents the reduction in the number of days over the course of a year that one experiences a given condition, such as shortness of breath, nausea, or headache. In a relatively small number of cases ($N = 17$), this variable represents a change in

Table 2. Descriptions of variables used in the meta-analysis

Variables	Description
WTPACUTE	Mean WTP for health change (in 2000 US dollars) ^a
ΔDAYS	Reduction in number of days with acute effect
ΔQWB	Improvement in health-related quality of life on affected days (= 1 – total QWB score)
QWBSYMSCORE	QWB symptom score (27 symptoms)
QWBMOBSCORE	QWB mobility score (3 levels)
QWBSACSCORE	QWB social activity score (3 levels)
QWBPACSCORE	QWB physical activity score (3 levels)
INCOME	Mean household income (in 2000 US dollars) ^a
AGE	Mean age
% MALE	Percent male
US	= 1 if study was conducted in the US; = 0 otherwise
WTPAVOID	= 1 if value was stated for avoiding a decrease in health; = 0 otherwise
OPEN ENDED	= 1 if an open-ended value elicitation method was used; = 0 otherwise
PAYMENT CARD	= 1 if a payment card value elicitation method was used; = 0 otherwise
IN PERSON	= 1 if survey was conducted as an in-person interview; = 0 otherwise
JOURNAL	= 1 if publication was published in a peer-reviewed journal; = 0 otherwise
SAMPLE SIZE	Number of respondents used to estimate the WTP value

^a Converted to dollars using the PPP index if in foreign currency; median WTP used if mean value not reported.

Table 3. Summary statistics for variables used in the meta-analysis

Variables	N	Mean	SD	Min	Median	Max
WTPACUTE	236	270.22	322.06	2.70	145.62	2927.69
ΔDAYS	236	11.9	20.5	1	5	90
ΔQWB	236	0.37	0.11	0.17	0.36	0.57
QWBSYMSCORE	235	0.26	0.03	0.17	0.257	0.30
QWBMOBSCORE	235	0.03	0.03	0	0	0.09
QWBSACSCORE	235	0.05	0.04	0	0.061	0.11
QWBPACSCORE	235	0.03	0.03	0	0	0.08
INCOME	236	46 348	13 618	21 891	47 067	88 020
AGE	236	45.2	6.8	35.4	44.5	68
% MALE	236	49.25	14.74	0	47.7	100
US	236	0.39	0.49	0	0	1
WTPAVOID	236	0.88	0.33	0	1	1
OPEN ENDED	236	0.14	0.35	0	0	1
PAYMENT CARD	236	0.29	0.46	0	0	1
IN PERSON	236	0.42	0.49	0	0	1
JOURNAL	236	0.67	0.47	0	1	1
SAMPLE SIZE	236	316.04	151.91	20	399	832

health that was expressed as a reduction in the number of acute events, such as asthma, angina, or allergy attacks. We assumed that each of these events was roughly equivalent to a day of acute illness, and in statistical tests of the regression results, the effect of ΔDAYS on WTP was not found to be significantly different for these observations.

Second, we used the QWB Scale to characterize the severity of the acute health effect. The QWB Scale characterizes health outcomes in four

dimensions – symptoms, mobility, social activity, and physical activity – each of which can be scored separately. Using the health effect descriptions from the valuation studies, we assigned each health effect to a defined level for all four dimensions. We then used the premeasured QWB weights [30] to assign a numerical score to each level.

The resulting four scores were captured in the variables QWBSYMSCORE, QWBMOBSCORE,

QWSACSCORE, and QWPACSCORE. For example, Johnson *et al.* [31] describe a condition as one day of 'coughing, wheezing, shortness of breath' and 'cannot leave your house, go to work, go to school, do housework, participate in social or recreational activities, and you have some physical limitations (trouble bending, stooping or doing vigorous activities) because of this health condition, but you can care for yourself.' Using the QWB system, this corresponds best with symptom category 11 (QWSYMSCORE = 0.257), mobility scale category 4 (QWMOBScore = 0.062), physical activity scale category 3 (QWPACSCORE = 0.060), and social activity scale category 1 (QWSACSCORE = 0.106) and $\Delta\text{DAYS} = 1$. Adding the individual QWB scores for this condition, the composite score becomes $\Delta\text{QWB} = 0.166$. Using this approach, a *higher* score in each of these cases indicates a more severe condition. As shown in Table 3, the symptom scores are on average higher than the other three, but the standard deviation of this score is slightly less. It should be noted that for one study [32], the health effect of interest could not be mapped into the four separate scores, but an overall QWB score was estimated in the study. As a result, one observation is missing for each of the four scores.

The variable ΔQWB provides a summary measure of the reduction in severity of the health effect. It was calculated as the sum of the four scores. This additive assumption is somewhat arbitrary, but it is consistent with the way a total QWB score is typically estimated for a specified health condition. The total QWB score is typically calculated as $1 - \text{QWSYMScore} - \text{QWMOBScore} - \text{QWSACScore} - \text{QWPACScore}$, such that a higher index represents better health. Therefore, ΔQWB is equal to $1 - \text{the total QWB score}$.

Expanding beyond the Johnson *et al.* approach, we also collected data from each study and included variables to control for other factors that might influence WTP estimates. The variables INCOME, AGE, %MALE, and US were included to account for potentially influential characteristics of the study population. Most studies report summary statistics for these characteristics, but this information is not always reported for the specific subsample that is used to calculate the WTP value. When the subsample information was not available, we used summary statistics for the full study sample.

The dummy variable WTPAVOID was specified to distinguish between WTP values that were

estimated for avoiding a decline in health, as opposed to those for improving health from current conditions. A positive effect for this variable would be consistent with declining marginal utility of health. However, a large positive effect for this variable may also indicate deviations from the standard utility model, such as the loss aversion and reference dependence models proposed by Tversky and Kahneman [26].

The remaining dummy variables were included to control for study design effects and for potential publication bias. Finally, we included SAMPLE SIZE, not as an explanatory variable, but as a weighting variable, so that WTP estimates based on larger samples could be given more weight in the regression analysis. The results of including these variables in meta-regressions are described below.

Other factors not specifically included in our list of variables may also influence WTP. For example, prices for medical care, wages (opportunity cost of sick time), and average education are all potentially influential factors. Unfortunately, the amount of information reported in the original studies is not adequate to include these factors in the analysis. However, the income variable may capture part of the wage effect, and the US dummy may serve as a proxy for differences in medical costs between the US and other countries.

Meta-regression models and results

Tables 4 and 5 describe the regression results for several model specifications, all of which share the same basic structure. We included a measure of WTP in all models as the dependent variable. Measures of the change in duration and severity of the corresponding health effect were included as explanatory variables. Additional explanatory variables include characteristics of the study population, valuation method, study design, and publication outlet.

Model specification

To explore the robustness of the results across model specifications, eight sets of results are reported in these tables. Two model specification issues in particular – functional form and health index specification – are addressed in the regres-

Table 4. Weighted least squares meta-regression results with total QWB score ($N = 236$)

Explanatory variable	Dependent variable: WTPACUTE				Dependent variable: ln(WTPACUTE)			
	Coefficient	t -stat ^a	Coefficient	t -stat ^a	Coefficient	t -stat ^a	Coefficient	t -stat ^a
Δ DAYS	4.78	3.59			0.02	4.36		
ln(Δ DAYS)			97.40	4.48			0.50	12.59
Δ QWB	1613.27	2.82			7.23	4.89		
ln(Δ QWB)			454.20	2.23			1.97	3.26
INCOME	0.00	0.57			0.00	1.82		
ln(INCOME)			197.20	1.61			0.70	2.13
AGE	-0.96	-0.05			0.09	2.20		
ln(AGE)			-58.22	-0.08			2.56	1.78
%MALE	-2.90	-0.70	-1.61	-0.58	-0.03	-2.05	-0.01	-1.36
US	18.87	0.14	-41.88	-0.38	-0.19	-0.64	-0.41	-1.48
WTPAVOID	-122.39	-0.34	-43.27	-0.15	0.75	0.99	0.78	1.33
OPEN ENDED	51.77	0.28	93.63	0.84	0.33	0.85	0.20	0.76
PAYMENT CARD	-15.06	-0.16	31.94	0.51	-0.06	-0.22	-0.02	-0.09
IN PERSON	-42.98	-0.33	-124.55	-1.08	-0.11	-0.32	-0.47	-1.28
JOURNAL	43.13	0.26	7.91	0.05	-0.54	-1.37	-0.71	-1.88
CONSTANT	-256.01	-0.22	-1168.54	-0.31	-1.75	-0.64	-10.34	-1.29
R^2 (%)	44.47		48.00		57.19		64.48	

^aBased on robust standard error estimates, corrected for clustering by study.

Table 5. Weighted least squares meta-regression results with four-dimensional QWB scores ($N = 235$)

Explanatory variable	Dependent variable: WTPACUTE				Dependent variable: ln(WTPACUTE)			
	Coefficient	t -stat ^a	Coefficient	t -stat ^a	Coefficient	t -stat ^a	Coefficient	t -stat ^a
Δ DAYS	4.66	3.70			0.02	4.67		
ln(Δ DAYS)			95.09	4.22			0.49	13.15
β^b			289.81	1.65			1.18	1.61
QWBSYSCORE	-88.88	-0.11	0.50 ^c	0.86	1.09	0.39	0.47 ^c	0.95
QWBMOSCORE	2896.72	5.86	2.78 ^c	1.34	10.94	9.93	2.73 ^c	3.92
QWBSACSCORE	880.54	1.15	0.31 ^c	0.93	4.88	1.24	0.45 ^c	1.24
QWBPACSCORE	2380.64	1.86			10.49	3.65		
INCOME	0.00	0.62			0.00	2.08		
ln(INCOME)			240.60	1.86			0.88	2.46
AGE	1.93	0.10			0.10	2.70		
ln(AGE)			52.72	0.07			3.04	2.16
%MALE	-5.47	-1.39	-3.87	-1.33	-0.04	-3.84	-0.03	-3.51
US	40.36	0.30	-24.52	-0.21	-0.12	-0.40	-0.35	-1.27
WTPAVOID	-205.62	-0.55	-65.84	-0.22	0.30	0.43	0.57	1.03
OPEN ENDED	40.00	0.22	121.37	1.13	0.33	0.98	0.33	1.48
PAYMENT CARD	10.67	0.12	63.47	0.99	0.08	0.29	0.14	0.68
IN PERSON	-14.55	-0.11	-119.09	-0.91	0.02	0.04	-0.43	-1.07
JOURNAL	44.73	0.29	19.41	0.12	-0.50	-1.31	-0.65	-1.79
CONSTANT	209.85	0.17	-1,978.06	-0.52	0.22	0.09	-13.58	-1.59
R^2 (%)	47.07		NA		59.46		NA	

^aBased on robust standard error estimates, corrected for clustering by study.

^bSee Equation (9) for interpretation of this coefficient.

^cRefers to η coefficient in Equation (9).

sion results tables. In addition, two model estimation issues – regression weights and clustering effects – were addressed in the meta-regressions and

are incorporated in the results tables. Each of these issues is described briefly below. It should also be noted that the robustness of the results described

in these tables is further confirmed by the fact that the size and statistical significance of the key variables change very little when the two highest WTP estimates (over \$1000 each) are excluded from the regression analysis.

Functional form: To evaluate the robustness of model results, we applied linear, semi-log, and log-linear specifications to analyze the data. Although, all these approaches are reasonable for approximating the relationship between WTP estimates and the other variables described in Table 2, the log-linear approach has a few conceptual advantages. First, it implies that, as changes in severity and duration of illness (and income) approach zero, WTP also approaches zero. Second, it implies that the marginal effect of income on WTP and the marginal effects of the severity and duration changes on WTP are not mutually independent. So, for example, the additional WTP associated with a larger health improvement is not assumed to be independent of an individual's budget. Third, as discussed in more detail below, the log-linear form allows for a more explicit statistical test of the QALY-based valuation approach.

Health index specification. All the results reported in these tables rely on the QWB Scale to describe the severity of acute illness. However, using the composite QWB score by itself implies that the four subcomponent scores – the symptoms, mobility, social activity, and physical activity scores – each have the same marginal effect on WTP. A less restrictive specification allows the four scores to enter the functional relationship separately and independently. In this way it is possible to test directly the simple additive assumption underlying the composite QWB score.

Regression weights: Treating each value estimate equally in the meta-regression is not efficient because it fails to account for the fact that some values are estimated with relatively more precision than others and therefore contribute more information to the analysis. To maximize efficiency, each estimate would ideally be weighted by the inverse of its variance; unfortunately relatively few of the included studies report standard errors or confidence intervals for their WTP estimates. As an approximation of these relative weights, we instead weighted each observation in direct proportion to its sample size (in contrast to the approach used by Johnson *et al.*, which used equal weighting).

Panel data modeling: To account for the panel nature of the data, we estimated all the models

using clustered robust regression [33]. Because the data used in the analysis are characterized by multiple observations from individual studies, they are likely to violate the ordinary least squares assumptions of independent and identically distributed errors. Therefore, we used the Huber–White method to correct the standard error estimates for potential error correlation within study clusters and unequal variance of errors across clusters [34,35].

Model results

The meta-regression results are summarized in Tables 4 and 5. The two sets of regressions differ primarily with respect to the health index specification.

Results with a composite QWB score. Table 4 reports results for linear, log-linear, and semi-log specifications of WTPACUTE with respect to Δ DAYS and the single composite QWB score (Δ QWB). All the models show a reasonably good fit, with R^2 -statistics between 45 and 65%. More importantly, several of the coefficients have the expected sign and are statistically significant.

The results indicate that the WTP estimates pass the scope test. The coefficients for Δ DAYS and Δ QWB, in both linear and log forms, are consistently positive and predominantly significant across specifications at a 0.05 level. In other words, average WTP to avoid acute effects increases with the number of days or acute episodes avoided and with the severity of the conditions avoided. The coefficients for the linear specification indicate that WTP increases by an average of \$4.80 for each additional day of acute illness avoided. Therefore, although statistically significant, the estimated marginal WTP for reduced duration of illness is quite small in the linear specification. Unlike the log-linear model, this specification does not restrict WTP to be zero when Δ DAYS is zero, nor does it allow marginal WTP to decline with respect to Δ DAYS. Consequently, marginal WTP may be underestimated for low levels of Δ DAYS. In this model, WTP is estimated to increase by \$16 for each 0.01 change (between 0 and 1) in the total QWB score. Using a linear specification implies that these marginal values for Δ DAYS and Δ QWB are constant and independent of one another; therefore, these estimates are best interpreted as average effects across the range of Δ DAYS and

ΔQWB . The log-linear specification discussed below relaxes these restrictions.

As expected, income also has a consistently positive and, for the most part, reasonably significant effect on WTP. Positive income effects support the hypothesis that health is a normal good. The income coefficient in the log-linear specification can be interpreted as an elasticity of WTP with respect to income. The elasticity estimate is 0.7, which implies that income has a positive but relatively inelastic effect on WTP. Notably, controlling for this income effect, the dummy variable for studies done in the United States does not have a significant effect (at a 0.05 level) on WTP in any of these specifications.

The average age of the sample also tends to have a positive effect, although the significance of this variable varies across specifications. To the extent that the age is an inverse proxy for health status, this result is consistent with declining marginal utility of health. In other words, this result suggests that older and less healthy individuals are willing to pay more for an increment in health. The estimated size of this age effect is surprisingly large, however. In the log-linear specification, the elasticity of WTP with respect to age is 2.56.

The coefficient on WTP_{AVOID} is generally not statistically significant. Its sign varies across specifications, although it is more often positive. Therefore, these results do not indicate the presence of loss aversion or strong reference effects.

None of the other variables characterizing the study approach show significant effects on WTP. However, in a few specifications, the JOURNAL coefficient is negative and significant at a 0.10 level, which suggests that values published in peer-reviewed journals are generally lower.

Results with a decomposition of the QWB score. Table 5 reports results for four similar specifications; however, the main difference is that the QWB score is decomposed into its four components.

For the most part, the results using the four QWB scores continue to show significant scope effects. The estimated coefficients for the scores are mostly positive and, particularly for the mobility score, are often statistically significant. For example, the results of the first specification in Table 5 indicate that average annual WTP increases by \$24 and \$29 for each 0.01 increment in the physical activity and mobility scores, respectively.

The decomposition of ΔQWB in these specifications also allows us to test one of the underlying assumptions of the QWB Scale (i.e. that each of the four separate dimensions of the index contributes equally to utility (and thus WTP)). In the linear specification and the semilog specification using $\ln(WTP)$ as the dependent variable ΔQWB is linearly decomposed. Separate coefficients are estimated for each of the four scores.

The results of the decomposed model indicate that the mobility score and the physical activity score have larger and consistently significant and positive effects on WTP. The symptom score and, to a lesser extent, the social activity score have lower and generally insignificant effects on WTP. It is worth reemphasizing that the regressions estimate the marginal effects of these scores on *private* WTP. If individuals are directly compensated for lost work time through sick leave policies, then the additional WTP of employers or society to avoid these conditions is not captured by these measures. If, in contrast, the costs of lost work time were to be fully internalized by the individuals experiencing the health effect, then it is likely that the mobility, physical activity, and social activity scores would have a larger effect on private WTP. It is also possible that the relative effects of these scores would be different (e.g. social activity restrictions might have a larger effect on WTP relative to physical activity restrictions).

This finding – that the different scores have different effects on individuals' utility and WTP – contradicts the implicit assumptions of the composite ΔQWB score. *F*-tests of the restriction that all four of the scores have the same marginal effect on WTP can be rejected at a 0.05 level for each of these specifications.

A comparable analysis of the other specifications is somewhat more complicated. In these cases, a linear decomposition of ΔQWB results in nonlinear models. For example, the simple log-linear model from Equation (7) can be rewritten as

$$\ln(WTP) = \beta \ln(\Delta QWB) + \beta \ln(\Delta DAYS/365) + \alpha X \quad (8)$$

where X represents the vector of all the other explanatory variables, which act as shifters on the intercept term. When ΔQWB is linearly

decomposed, it results in the following nonlinear form:

$$\begin{aligned} \ln(\text{WTP}) = & \beta \ln(\eta_1 \text{QWBSYMSCORE} \\ & + \eta_2 \text{QWBMOBScore} \\ & + \eta_3 \text{QWBSACScore} \\ & + \eta_4 \text{QWBPAcScore}) \\ & + \beta \ln(\Delta \text{DAYS}/365) + \alpha X \end{aligned} \quad (9)$$

We estimated the nonlinear models with the QWB subcomponent scores using maximum likelihood estimation and report the results in Table 5. In each of these specifications, one of the η coefficients (in this case η_4) is restricted to be equal to one. This restriction implies that, if the marginal effects of the four scores on WTP are the same, then the other three η coefficients will also equal one. Furthermore, if these coefficients are all equal to one, then the specification in Equation (9) simplifies to Equation (8).

The results of these nonlinear specifications again cast doubt on the assumption underlying the composite QWB score. The coefficients on the mobility and physical activity scores are distinctly larger than for the other two QWB scores. To test the restrictions implied by using the sum of the four scores (ΔQWB) as a composite index, in these cases we can compare the log-likelihood values for the restricted models (in Table 4) with those for the unrestricted models (in Table 5). Based on likelihood ratio tests, the restricted models can all be rejected at a 0.05 level of significance.

In the models reported in Table 5, the effects of duration income, age, and other variables are, for the most part, consistent with the previous results. The coefficients on ΔDAYS (linear and log form) are virtually unchanged and consistently positive and significant. The effect of $\ln(\text{INCOME})$ is also consistently positive and significant, although the coefficient is somewhat larger. The final specification in Table 5 implies an income elasticity of almost 0.9. The estimated elasticity of WTP with respect to age continues to be positive and significant, and once again it is surprisingly large (i.e. greater than 3 in Table 5). The effect of gender composition (%MALE) continues to be negative but is somewhat larger (in absolute value) and more significant.

Testing the QALY valuation assumptions. The results of the log-linear specifications in both Tables 4 and 5 can also be used to test the assumptions of

the QALY valuation approach for estimating morbidity values. The results do not support these assumptions.

According to the QALY valuation approach, WTP is assumed to increase in direct proportion to the gain in QALYs. As previously discussed, it is often assumed that the value per QALY is approximately \$100 000. Under less restrictive assumptions, the relationship between WTP and QALYs can be expressed as

$$\text{WTP} = \alpha(\Delta \text{QALY})^\beta \quad (6')$$

In other words, the QALY valuation approach restricts β to be equal to one. If β is greater than or equal to one, then WTP will increase more or less than proportionately with respect to QALY gains.

If we assume that the QWB Scale is an appropriate health utility index, and we assume that the QALY gain of interest is brought about by avoiding a health effect of specific duration and severity, then Equation (6') can be expanded as follows:

$$\text{WTP} = \alpha(\Delta \text{DAYS}/365 \times \Delta \text{QWB})^\beta \quad (10)$$

In this expression, the QALY gain is expressed as the constant per-period utility gain (from less than 1 to 1) times the duration of the utility gain (converted to years). In log-linear form, this equation becomes

$$\begin{aligned} \ln(\text{WTP}) = & \ln(\alpha) + \beta \ln(1/365) \\ & + \beta \ln(\Delta \text{DAYS}) + \beta \ln(\Delta \text{QWB}) \end{aligned} \quad (11)$$

This expression implies that, with a log-linear model, the restrictions imposed by the QALY valuation approach can be tested directly. The null hypothesis is that coefficients on $\ln(\Delta \text{DAYS})$ and $\ln(\Delta \text{QWB})$ are both equal to one. An F-test of this restriction for the log-linear specification in Table 4 can be strongly rejected (p -value < 0.001). In the last specification, the coefficient estimate for $\ln(\Delta \text{DAYS})$ is considerably less than one (0.5), and the coefficient estimate for $\ln(\Delta \text{QWB})$ is considerably greater than one (1.97). Therefore, even the less restrictive assumptions implied by Equation (6') – that coefficients on $\ln(\Delta \text{DAYS})$ and $\ln(\Delta \text{QWB})$ are equal to one another (but not necessarily equal to one) – can be strongly rejected (p -value = 0.026).

The results of the log-linear specification indicate that WTP increases less than proportionately with respect to changes in duration, as measured by ΔDAYS . This result is consistent with a declining marginal disutility with respect to duration

of illness. Furthermore, the coefficient for $\ln(\Delta QWB)$ implies that WTP increases more than proportionately with the severity of illness. This result is consistent with a decreasing marginal utility with respect to health status.

The final (log-linear) specification in Table 5 can also be used to test the restrictions implied by the QALY valuation approach. By replacing ΔQWB in Equation (11) with a reweighted composite score based on the results in Table 5 ($\eta_1 QWBSYMSCORE + \eta_2 QWBMOBSCORE + \eta_3 QWBSACSCORE + 1 QWBPACSCORE$), it is again possible to test the restriction that the coefficient for $\ln(DAYS)$ is equal to the β parameter. Using a likelihood ratio test, this restriction can again be rejected at a 0.05 level of significance.

Implications for policy analysis

The overall goodness of fit and statistical significance of the models reported in Tables 4 and 5 suggest that they provide a reasonable foundation for constructing a predictive WTP function. In

other words, the models can be used to develop and test alternative benefit transfer functions.

The advantage of being able to specify a WTP function is that it can be used to extrapolate beyond the existing set of WTP estimates in the literature. That is, it can be used to estimate values for any number of avoided acute illnesses, as long as these illnesses can be described according to their severity (in this case, using the QWB classification system) and their duration. In addition, depending on the model specification and econometric results, it can be used to tailor WTP estimates according to the socio-demographic characteristics of the affected population (e.g. average age, gender composition, and/or average income). Developing value estimates that vary according to income levels can raise difficult ethical issues. Nonetheless, it may be useful, for example, to estimate WTP for a person with average income or to estimate how average WTP will be affected by growth in per-capita income levels.

To demonstrate and evaluate how the results can be used to predict WTP for avoiding acute effects under specific conditions, we selected two specifications – the log-linear specification using the composite QWB score (last specification in Table 4) and the log-linear form using the four

Table 6. Benefit transfer function estimates

Explanatory variable	BT function 1 ^a		Explanatory variable	BT function 2 ^b	
	Coefficient	<i>t</i> -stat ^c		Coefficient	<i>t</i> -stat ^c
$\ln(\Delta DAYS)$	0.501	13.07	$\ln(\Delta DAYS)$	0.477	12.93
$\ln(\Delta QWB)$	2.339	6.23	β^d	1.239	2.22
			$QWBSYMSCORE (\eta_1)^d$	0.357	1.31
			$QWBMOBSCORE (\eta_2)^d$	2.120	4.45
			$QWBSACSCORE (\eta_3)^d$	0.476	2.22
			$QWBPACSCORE (\eta_4)^d$	1.000	
$\ln(INCOME)$	0.777	3.01	$\ln(INCOME)$	0.833	3.96
$\ln(AGE)$	2.591	2.06	$\ln(AGE)$	2.987	2.49
US	-0.181	-1.52	%MALE	-0.017	-2.61
WTPAVOID	0.799	1.72	US	-0.101	-0.73
JOURNAL	-0.357	-2.05	WTPAVOID	0.516	1.25
CONSTANT	-12.031	-1.66	JOURNAL	-0.382	-1.96
			CONSTANT	-13.293	-2.02
R^2	63.50%			NA	
<i>N</i>	236			235	

^a Estimated with weighted least squares regression (SAMPLE SIZE weight).

^b Estimated with weighted maximum likelihood regression (SAMPLE SIZE weight).

^c Based on robust standard error estimates, corrected for clustering by study.

^d See Equation (9) for interpretation of coefficient.

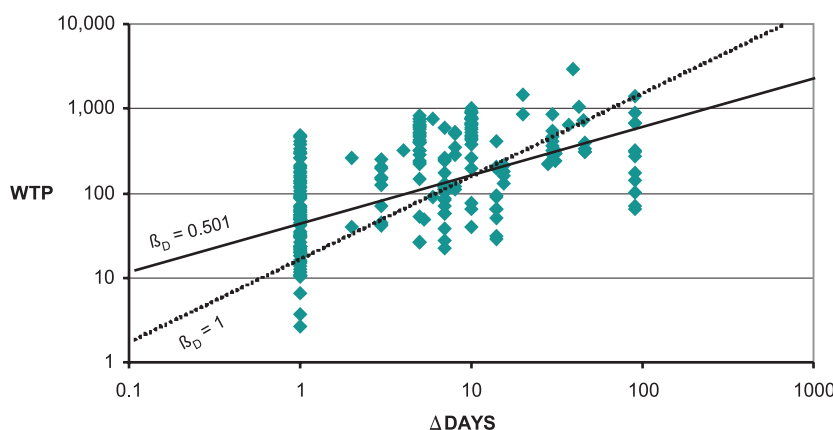


Figure 1. Observed and fitted WTP estimates with respect to duration of illness $^a\beta_D = \partial \ln(\text{WTP}) / \partial \ln(\Delta \text{DAYS})$

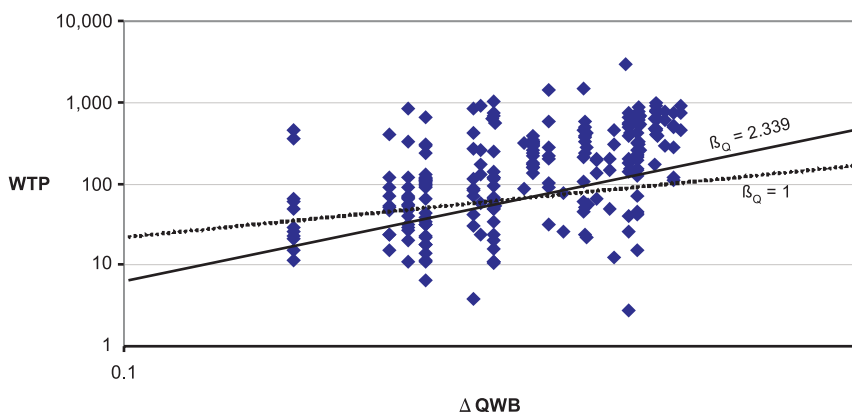


Figure 2. Observed and fitted WTP estimates with respect to severity of illness $^a\beta_Q = \partial \ln(\text{WTP}) / \partial \ln(\Delta \text{QWB})$

separate scores (last specification in Table 5). We selected these specifications because, as argued previously, they are most defensible from an *a priori* standpoint and because they provide strong empirical results in terms of goodness of fit and statistical significance.

In both cases, we conducted specification tests and dropped the least significant variables. The revised specifications are reported in Table 6 as BT Function 1 and BT Function 2, respectively. For BT Function 1, the three variables related to survey method (OPEN ENDED, PAYMENT CARD, and IN PERSON) and the gender proportion variable (%MALE) were dropped from the original specification. Based on an *F*-test, the coefficients for these four variables were jointly not significantly different from zero. For BT

Function 2, only the three survey method variables were dropped from the original specification. Using a likelihood ratio test, a restricted model that also held the %MALE coefficient at zero could not be rejected at a 0.05 level of significance.

Before evaluating the out-of-sample implications of these models, it is informative to compare the fitted model with the estimation sample. Figures 1 and 2 plot the 236 WTP estimates used to estimate BT Function 1 against their respective ΔDAYS and ΔQWB values. Although there is some clustering of ΔDAYS and ΔQWB on specific values, the plots show a reasonably good dispersion of data in both cases (on the logarithmic scale). There is also a loose but notable upward trend in WTP with respect to both ΔDAYS and ΔQWB . Both figures also include a fitted

regression line based on the parameter estimates for BT Function 1. The dark line in Figure 1 shows the model predicted value for $\ln(\text{WTP})$ with respect to $\ln(\Delta\text{DAYS})$, holding all other explanatory variables at their respective means. For ease of interpretation, $\ln(\Delta\text{DAYS})$ and $\ln(\text{WTP})$ are shown without logarithmic transformation on their respective axes. A similar approach is used in Figure 2 for $\ln(\text{WTP})$ and $\ln(\Delta\text{QWB})$. Both of these lines can be compared to the dotted line, whose slope represents unit elasticity (i.e. the assumed elasticity under the QALY valuation approach). The fitted slope of $\ln(\text{WTP})$ with respect to $\ln(\Delta\text{DAYS})$ is clearly less than unit elasticity, and the opposite is true for the fitted slope of $\ln(\text{WTP})$ with respect to $\ln(\Delta\text{QWB})$.

To explore the implications of the BT functions under selected out-of-sample conditions, we applied each function to predict average individual WTP for four selected acute conditions. As shown in Table 7, each of the selected acute conditions corresponds to a different severity description, which is defined using the QWB health dimensions and scores. The symptom descriptions vary from relatively minor ('breathing unpleasant air') to relatively severe ('sick, vomiting'), and the three functional restrictions – mobility, physical activity, and social activity – are alternatively moderate or severe. For each of the four conditions, the duration of illness avoided (ΔDAYS) is also evaluated at either 1 or 10 days.

For both BT Function 1 and BT Function 2, Table 7 shows the mean predicted WTP and corresponding 95% confidence interval (CI) for each condition and duration (eight combinations). Each predicted value is estimated using the specified duration and QWB scores. The values of the other explanatory variables are held constant across the eight scenarios. In all cases, INCOME is set to \$45 000 per year, AGE is set at 45 years, and for BT Function 2 the %MALE is 50%. For these illustrations of the BT functions, the values are also all estimated assuming a US population ($\text{US} = 1$) and for avoiding health declines ($\text{WTP}_{\text{AVOID}} = 1$). Furthermore, assuming that values published in peer-reviewed articles are most defensible, we set JOURNAL equal to one as well. To appropriately transform the model prediction of mean $\ln(\text{WTP})$ to mean WTP in nonlogarithmic form, we applied a smearing factor to the model-predicted values [36]. The smearing factor is equal to the average of the exponentiated residuals for the observations in the estimation sample.

For BT Function 1, the predicted mean WTP values range from \$45 (95% CI: \$39–\$51) for avoiding one day, poor air and moderate functional restrictions to \$706 (95% CI: \$345–\$1067) for avoiding 10 days of sickness and vomiting with severe functional restrictions. For each condition, the 10-fold increase in duration has a positive but closer to 3-fold effect on predicted WTP. In contrast, increasing total ΔQWB by a factor of two (between conditions 1 and 2) increases predicted WTP substantially, by a factor of roughly 5.

For BT Function 2, the predicted mean WTP values show less variation. They range from \$124 (95% CI: \$68–\$181) to \$689 (95% CI: \$227–\$1150). As expected, based on the parameter estimates for this function, predicted WTP is somewhat less sensitive to differences in the duration of illness. Predicted WTP is also less sensitive to variations in severity using BT Function 2. For the least severe condition, predicted WTP is nearly three times as large as the WTP estimate based on BT Function 1; however, it is roughly the same of the most severe condition. In particular, BT Function 2 is relatively less sensitive to changes in the QWB symptom scores (holding the functional restriction scores constant, as between conditions 1 and 2 and between conditions 3 and 4).

For comparative purposes, Table 6 also includes value estimates based on a simple QALY valuation approach (using QWB as the health utility index) for each scenario. In five of the eight scenarios, these QALY-based estimates are higher than one or both of the BT function estimates. They are particularly large compared to the BT function estimates when the duration change is large (e.g. 10 days) and/or when the severity of illness is lower. These differences are consistent with the different slopes shown in Figures 1 and 2. They occur because the QALY-based approach assumes that values increase in direct proportion to duration and to severity, whereas the BT functions have a lower elasticity (< 1) with respect to duration and a larger elasticity (> 1) with respect to severity.

Conclusions

Given the increasing importance placed on CBA for evaluating public health and safety policies, policy makers need to be able to consistently

Table 7. Illustrative out-of-sample WTP predictions

<i>Description</i>	Acute condition 1			Acute condition 2			Acute condition 3			Acute condition 4		
	1	10		1	10		1	10		1	10	
Duration ΔDAYS												
Severity: symptoms	Breathing smog or unpleasant air			Cough, wheezing, shortness of breath			Headache or dizziness			Sick, vomiting		
QWBSYSCORE	0.101	0.101		0.257	0.257		0.244	0.244		0.29	0.29	
Severity: functional restrictions												
	Moderate			Moderate			Severe			Severe		
QWBMOBSORE	0.062	0.062		0.062	0.062		0.09	0.09		0.09	0.09	
QWBSACSCORE	0.06	0.06		0.06	0.06		0.077	0.077		0.077	0.077	
QWBPACSCORE	0.061	0.061		0.061	0.061		0.106	0.106		0.106	0.106	
Severity: total ΔQWB	0.284	0.284		0.44	0.44		0.517	0.517		0.563	0.563	
<i>Valuation</i>												
BT function 1 ^{a,b}												
95% CI Lower	\$38.78	\$129.94		\$88.59	\$264.25		\$109.01	\$318.47		\$119.68	\$345.19	
Predicted WTP/case avoided	\$44.94	\$142.51		\$125.12	\$396.73		\$182.43	\$578.49		\$222.68	\$706.10	
95% CI Upper	\$51.10	\$155.07		\$161.64	\$529.22		\$255.86	\$838.51		\$325.68	\$1067.02	
BT function 2 ^{a,b}												
95% CI Lower	\$67.54	\$227.59		\$100.26	\$257.74		\$110.91	\$263.63		\$102.08	\$226.85	
Predicted WTP/case avoided	\$124.11	\$371.86		\$158.32	\$474.39		\$218.80	\$655.58		\$229.83	\$688.66	
95% CI Upper	\$180.67	\$516.13		\$216.39	\$691.04		\$326.68	\$1047.53		\$357.58	\$1150.46	
QALY valuation estimate ^c	\$77.81	\$778.08		\$120.55	\$1205.48		\$141.64	\$1416.44		\$154.25	\$1542.47	

^a Confidence intervals calculated using the estimated asymptotic variance of the WTP functions.^b All WTP predictions assume the following values for the other explanatory variables: INCOME = \$45 000, %MALE = 50, AGE = 45, US = 1, WTPAVOID = 1, JOURNAL = 1.^c (ΔDAYS/365) (ΔQWB) × \$100 000.

estimate WTP for avoided illness. New primary research in health valuation is always desirable, but as this study shows, much can be learned from the existing body of empirical research.

The results of the meta-regression analysis indicate that WTP estimates for avoided acute effects vary in systematic and expected ways with respect to key explanatory variables. We find a strong statistical relationship between the value estimates and corresponding measures of the severity (using the QWB Scale) and duration of the health effects. We find that the average income and age of the study populations are statistically and economically significant explanatory factors in explaining WTP variation, which provides support for the theoretical validity of the stated preference approach. These results have also been confirmed by a recent study that expanded on the Johnson *et al.* meta-analysis with a somewhat different and less expanded sample of WTP estimates [37].

In addition, our meta-analysis results provide a simple but informative test of the assumptions underlying the QALY valuation approach for assessing morbidity values. The results strongly reject the assumption of a constant value per QALY and the assumption that the duration and the severity of illness have equivalent and proportional effects on WTP.

Finally, the results also indicate that the four health utility scores underlying the QWB Scale have statistically different effects on WTP. The mobility and physical activity dimensions were found to have stronger effects on WTP than the symptoms or social activity dimensions. This finding contradicts the equal weighting assumption, which is typically used in constructing the composite QWB health status index.

A key issue is therefore what these results imply for benefit transfer. First, the results cast doubt on the QALY valuation approach for predicting and transferring health benefits, at least for acute conditions. This result is consistent with the findings of others who have been critical of the approach, particularly for valuing changes in acute illness [38]. Second, the strong explanatory power, robustness, and statistical significance of the meta-regressions provide general support for their use as benefit transfer functions for policy analysis. In the absence of WTP studies that specifically match the policy effects of interest, these equations provide a systematic and empirically based approach for predicting WTP. BT Function 2 in particular provides a transfer tool that can differ-

entiate values according to the underlying dimensions of the health condition (i.e. symptoms, mobility, and activity).

Some caution is required, however, in interpreting and using the meta-analysis results. First, and most importantly, they only apply to acute conditions. Through our search of the health valuation literature, we determined that the number of WTP studies for chronic conditions is currently not adequate to support a comparable meta-analysis of these values. For chronic conditions, there are additional challenges in developing appropriate measures of duration and severity of illness. Second, the size of the estimated marginal effect of age on WTP in the meta-regressions, although statistically significant and reasonably robust across functional specifications, is unexpectedly high. Further investigation of the age effect on WTP is warranted, particularly as more WTP studies become available.

Third, it is important to emphasize that the meta-analysis function only includes estimates of adults' *private* WTP to avoid morbidity outcomes. As a result, it may not include all the value components that are of potential interest to policy makers. To the extent that costs of illness are externalized to other members of society, for example through social insurance, this analysis will not capture the benefits of avoiding these externalized costs. By excluding these externalized costs, estimates of *private* WTP may in some cases significantly underestimate the total social benefits of avoided illness. Moreover, the meta-analytic functions cannot be applied to *specifically* estimate values for protecting children's health because, by design, none of the included WTP estimates address altruistic (i.e. parental) values.

Fourth, it should be emphasized that the strength of these results depends in large part on the strength of the underlying WTP estimates and QWB scores. It is encouraging that the convergent validity of the two measures is supported by the finding that WTP is positively and significantly related to QWB scores. Nevertheless, the results would be weakened if either of these two preference measures includes systematic biases. Although the empirical accuracy and reliability of both methods have been questioned (see [39] for a critique of CVM methods and [16] for a review of QWB evaluations), there is no conclusive evidence that such systematic biases exist.

Table A1. Summary of studies included in the meta-analysis of WTP estimates^a

Study number	Study name	Valuation method	Health effect/change	Country	Number of WTP Obs
1	Alberini and Krupnick [40]	CVM	Acute respiratory symptoms	Taiwan	3
1	Alberini <i>et al.</i> [41]	CVM	Acute respiratory symptoms	Taiwan	2
2	Bala <i>et al.</i> [42]	CVM	Pain from shingles	US	1
3	Chestnut <i>et al.</i> [43]	CVM	Angina attacks	US	2
3	Rowe and Chestnut [44]	CVM	Asthma attacks	US	1
4	Dickie and Ulerly [45]	CVM	Acute symptoms	US	15
5	Johnson <i>et al.</i> [31]	Conjoint	Acute symptoms/activity restrictions	Canada	67
6	Kartman <i>et al.</i> [46]	CVM	Angina attacks	Sweden	12
7	Keith <i>et al.</i> [47]	CVM	Allergic rhinitis	US	1
8	Liu <i>et al.</i> [32]	CVM	Cold	Taiwan	1
9	Ready <i>et al.</i> [48]	CVM	Acute symptoms	Norway	10
9	Ready <i>et al.</i> [49]	CVM	Acute symptoms	Norway, Netherlands, Portugal, Spain, England	27
10	Navrud [50]	CVM	Acute symptoms	Norway	18
11	Lee <i>et al.</i> [51]	Conjoint	One day of influenza symptoms	US	3
12	Torrance <i>et al.</i> [52]	CVM	Acute exacerbation of chronic bronchitis	Canada	2
13	Jacobs <i>et al.</i> [53]	CVM	Hepatitis A	US	1
14	Chestnut <i>et al.</i> [54]	CVM	Angina attacks	US	2
15	Dickie <i>et al.</i>^b	CVM	Acute symptoms	US	12
16	Loehman <i>et al.</i> [55]	CVM	Acute symptoms	US	36
17	Tolley and Babcock [56]	CVM	Acute symptoms	US	20
				Total	236

^aPublications shown in bold font were included in the Johnson *et al.* [3] meta-analysis.^bDickie M, Gerking S, McClelland G, Schulze W. Contingent valuation: the value of formation process. Unpublished manuscript, 1988.

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Appendix A

The detailed descriptions of some of the key characteristics of these studies, including the valuation methods used, the health effects valued, the country studied, and the number of value estimates included in the meta-analysis are provided in Table A1.

References

1. Agency for Healthcare Research and Quality (AHRQ). Focus on cost-effectiveness analysis at the Agency for Healthcare Research and Quality. Fact Sheet. AHRQ Pub. No. 01-PO23, 2001.
2. Harvard School of Public Health (HSPH), Harvard Center for Risk Analysis. Catalog of preference scores. *The CEA Registry: Standardizing the Methods and Practices of Cost-Effectiveness Analysis*. Harvard Center for Risk Analysis, Harvard School of Public Health: Boston, MA. <http://www.hsph.harvard.edu/cearegistry/> or <http://www.hcra.harvard.edu/pdf/preferencescores.pdf> [7 November 2002].
3. Johnson FR, Fries EE, Banzhaf HS. Valuing morbidity: an integration of the willingness-to-pay and health-status index literatures. *J Health Econ* 1997; **16**: 641–665.
4. Cutler DM, Richardson E. *Measuring the health of the US population*. Brookings Papers on Economic Activity: Microeconomics. Brookings Institute: Washington, DC, 1997.
5. Zarkin GA, Dean N, Mauskopf JA, Williams Jr R. Potential health benefits of nutrition label changes. *Am J Public Health* 1993; **83**(5): 717–724.
6. Stanley TD. Wheat from chaff: meta-analysis as quantitative literature review. *J Econ Perspect* 2001; **15**(Summer): 131–150.
7. Smith VK, Pattanayak SK. Is meta-analysis the Noah's Ark for non market valuation? *Environ Resour Econ* 2002; **22**(1–2): 271–296.
8. Smith VK, Osborne LL. Do contingent valuation estimates pass a 'scope' test? A meta-analysis. *J Environ Econ Manage* 1996; **31**(November): 287–301.
9. Hanemann WM. Willingness to pay and willingness to accept: how much can they differ? *Am Econ Rev* 1991; **81**(3): 635–647.
10. Johannsson P-O. *Evaluating Health Risks: An Economic Approach*. Cambridge University Press: Cambridge, MA, 1995.
11. Harrington W, Portney PR. Valuing the benefits of health and safety regulation. *J Urban Econ* 1987; **22**: 101–112.
12. McConnell KE. Does altruism undermine existence value? *J Environ Econ Manage* 1997; **32**: 22–37.
13. Johannsson P-O. Altruism and the value of statistical life. *J Health Econ* 1994; **13**: 111–118.
14. Bergstrom TC. When is a man's life worth more than his human capital? In *The Value of Life and Safety*, Jones-Lee MW (ed.), North-Holland: New York, 1992; 3–26.
15. Johannesson M. *Theory and Methods of Economic Evaluation of Health Care*. Kluwer Academic Publishers: Dordrecht, The Netherlands, 1996.
16. Brazier J, Deverill M, Green C, Harper R, Booth A. A review of the use of health status measures in economic evaluation. *Health Technol Assess* 1999; **3**(9): 1–158.
17. Bleichrodt H, Quiggin J. When is cost-effectiveness analysis equivalent to cost-benefit analysis? *J Health Econ* 1999; **18**: 681–708.
18. Viscusi WK. The value of risks to life and health. *J Econ Lit* 1993; **31**: 1912–1946.
19. Klose T. A utility-theoretic model for QALYs and willingness to pay. *Health Econ* 2003; **12**(1): 17–31.
20. Carson RT. Contingent valuation: a user's guide. *Environ Sci Technol* 2000; **34**: 1413–1418.
21. Smith RD. The discrete-choice willingness-to-pay question format in health economics: should we adopt environmental guidelines? *Med Decis Making* 2000; **20**: 194–206.
22. Blamey RJ, Bennett W, Morrison MD. Yea-saying in contingent valuation surveys. *Land Econ* 1999; **75**: 126–141.
23. Ryan M, Scott DA, Donaldson C. Valuing health care using willingness to pay: a comparison of the payment card and dichotomous choice methods. *J Health Econ* 2004; **23**: 237–258.
24. Frew EJ, Whynes DK, Wolstenholme JL. Eliciting willingness to pay: comparing closed-ended with open-ended and payment scale formats. *Med Decis Making* 2003; **23**(2): 150–159.
25. Holmes TP, Kramer RA. An independent sample test of yea-saying and starting point bias in dichot-

- omous-choice contingent valuation. *J Environ Econ Manage* 1995; **29**: 121–132.
26. Tversky A, Kahneman D. Loss aversion in riskless choice: a reference-dependent model. *Q J Econ* 1991; **106**: 1039–1061.
 27. RTI. *Valuation of morbidity losses: meta-analysis of willingness-to-pay and health status measures*. Final Report, US Food and Drug Administration, RTI International: Research Triangle Park, NC, 2003.
 28. Diener A, O'Brien B, Gafni A. Health care contingent valuation studies: a review and classification of the literature. *Health Econ* 1998; **7**: 313–326.
 29. Olsen JA, Smith RD. Theory versus practice: a review of 'willingness-to-pay' in health and health care. *Health Econ* 2001; **10**: 39–52.
 30. Kaplan RM, Anderson JP, Wu AW, Mathews WC, Kozin F, Orenstein D. The quality of well-being scale: applications in AIDS, cystic fibrosis, and arthritis. *Med Care* 1989; **27**(3 Suppl.): S27–S43.
 31. Johnson FR, Banzhaf MR, Desvousges WH. Willingness to pay for improved respiratory and cardiovascular health: a multiple-format stated-preference approach. *Health Econ* 2000; **9**: 295–317.
 32. Liu J-T, Hammitt JK, Wang J-D, Liu J-L. Mother's WTP for her own and her child's health: a contingent valuation study in Taiwan. *Health Econ* 2000; **9**: 319–326.
 33. Rogers WH. Regression standard errors in clustered samples. *Stata Tech Bull* **13**: 19–23. Reprinted in *Stata Techn Bull Reprints* 1993; **3**: 88–94.
 34. Williams RL. A note on robust variance estimation for cluster-correlated data. *Biometrics* 2000; **56**: 645–646.
 35. Froot KA. Consistent covariance matrix estimation with cross-sectional dependence and heteroskedasticity in financial data. *J Finan Quant Anal* 1989; **24**: 333–355.
 36. Duan N. Smearing estimate: a nonparametric retransformation method. *J Am Stat Assoc* 1983; **78**(383): 605–610.
 37. Vassanadumrongdee S, Shunji M, Hiroaki S. Meta-analysis of contingent valuation studies on air pollution-related morbidity risks. *Environ Econ Policy Stud* 2004; **6**: 11–47.
 38. Bala MV, Zarkin GA. Are QALYs an appropriate measure for valuing morbidity in acute diseases? *Health Econ* 2000; **9**(2): 177–180.
 39. Hausman JA. *Contingent Valuation: A Critical Assessment*. Elsevier Science B.V.: Amsterdam, 1993.
 40. Alberini A, Krupnick A. Air quality and episodes of acute respiratory illness in Taiwan cities: evidence from survey data. *J Urban Econ* 1998; **44**(1): 68–92.
 41. Alberini A, Cropper M, Fu T-T *et al.* Valuing health effects of air pollution in developing countries: the case of Taiwan. *J Environ Econ Manage* 1997; **34**: 107–126.
 42. Bala MV, Wood LL, Zarkin GA, Norton EC, Gafni A, O'Brien B. Valuing outcomes in health care: a comparison of willingness to pay and quality-adjusted life-years. *J Clin Epidemiol* 1998; **51**(8): 667–676.
 43. Chestnut LG, Keller LR, Lambert WE, Rowe RD. Measuring heart patients' willingness to pay for changes in angina symptoms. *Med Decis Making* 1996; **16**: 65–77.
 44. Rowe RD, Chestnut LG. *Oxidants and asthmatics in Los Angeles: a benefits analysis*. Final Report, US Environmental Protection Agency. Document No. EPA/230/7-85/010, 1985.
 45. Dickie M, Ulery V. *Parental altruism and the value of child health: are kids worth more than parents?* Report, US Environmental Protection Agency, 2002.
 46. Kartman B, Andersson F, Johannesson M. Willingness to pay for reductions in angina pectoris attacks. *Med Decis Making* 1996; **16**(3): 248–253.
 47. Keith PL, Haddon J, Birch A, Rhinocort Study Group. A cost–benefit analysis using a WTP questionnaire of intranasal budesonide for seasonal allergic rhinitis. *Ann Allergy Asthma Immunol* 2000; **84**: 55–62.
 48. Ready RC, Navrud S, Dubourg WR. How do respondents with uncertain willingness to pay answer contingent valuation questions? *Land Econ* 2001; **77**(3): 315–326.
 49. Ready R, Navrud S, Day B *et al.* *Benefit transfer in Europe: are values consistent across countries?* Working Paper, 1999.
 50. Navrud S. Valuing health impacts from air pollution in Europe. *Environ Res Econ* 2001; **20**: 305–329.
 51. Lee PY, Matchar D, Clements D, Huber J, Hamilton J, Peterson E. Economic analysis of influenza vaccination and antiviral treatment for healthy working adults. *Ann Intern Med* 2002; **137**(4): 225–231.
 52. Torrance G, Walker V, Grossman R *et al.* Economic evaluation of ciprofloxacin compared with the usual antibacterial care for the treatment of acute exacerbations of chronic bronchitis in patients followed for 1 year. *Pharmacoeconomics* 1999; **16**(5 Pt. 1): 499–520.
 53. Jacobs RJ, Moleski RJ, Meyerhoff AS. Valuation of symptomatic hepatitis A in adults. *Pharmacoeconomics* 2002; **20**(11): 739–747.
 54. Chestnut LG, Colome SD, Keller LR *et al.* *Heart disease patients' averting behavior, costs of illness, and willingness to pay to avoid angina episodes*. Final Report, US Environmental Protection Agency. Document No. EPA-230-10-88-042, 1988.
 55. Loehman ET, Berg SV, Arroyo AA *et al.* Distributional analysis of regional benefits and cost of air quality control. *J Environ Econ Manage* 1979; **6**: 222–243.
 56. Tolley G, Babcock L. *Valuation of reductions in human health symptoms and risk*, in: vol. 3: *contingent valuation study of light symptoms and angina*. Final Report, US Environmental Protection Agency, 1986.